Supplementary Materials: Gradient-Sensitive Functional Descriptors for Interferometric Surface Metrology

1. Artefact Documentation

1.1. Artefact photographs

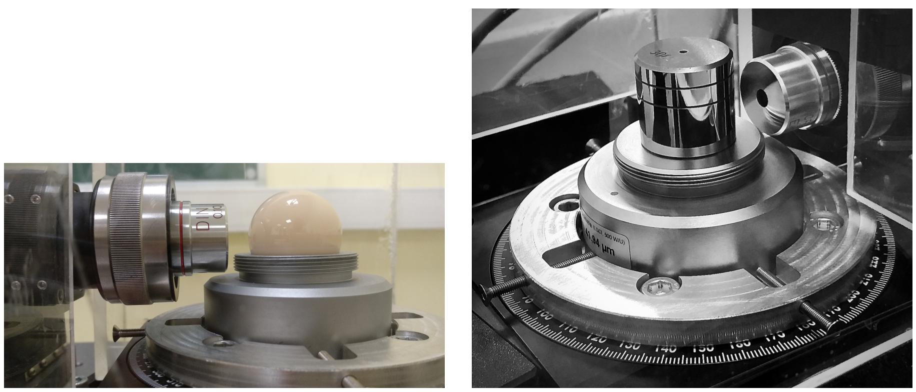

Figure S1: (a) JENOPTIK roundness standard FN 111 (Al_2O_3, $\phi = 29.9588$ mm), test report Art. 521 799, DAkkS-DKD calibration certificate Art. 532 529. (b) JENOPTIK magnification standard FN 101, test report Art. 521 809, DAkkS-DKD calibration certificate Art. 532 528.

2. Experimental Setup

2.1. Rotation stage specifications

Table S1: Technical specifications of the rotation stage [1].

Parameter	Specification
Angular range	Unrestricted number of revolutions
Maximum speed	80°/s
Encoder accuracy	2.0 mdeg
Wobble, typical	± 12 μrad
Wobble, guaranteed	± 25 μrad
Eccentricity, typical	± 0.40 μm
Transversal compliance	10 μrad/N m
Origin repeatability	± 0.25 mdeg

2.2. Environmental shielding and vibration isolation

Figure S2: (a) Setup shielded by a plexiglass box and covered by a wool blanket. (b) Layered wooden-polystyrene support with granite plate.

3. Image Processing — Phase Extraction

3.1. Single-image radial phase extraction

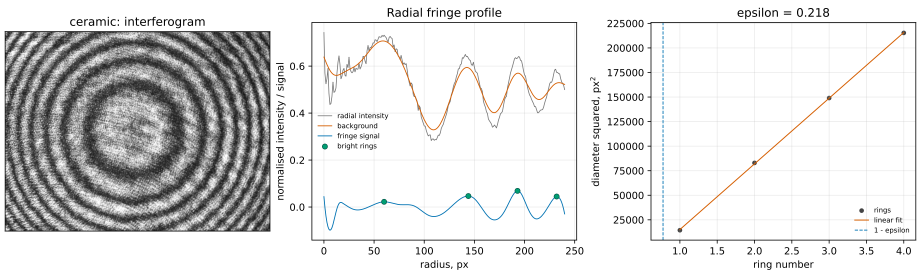

Figure S3: Ceramic single-image radial phase extraction. (Left) Raw interferogram (648×488 px). (Centre) Radial fringe intensity after background subtraction and B-spline smoothing with detected bright-ring maxima marked. (Right) D_j^2–order linear fit: the $D^2 = 0$ intercept (dashed) gives $1 - \epsilon$.

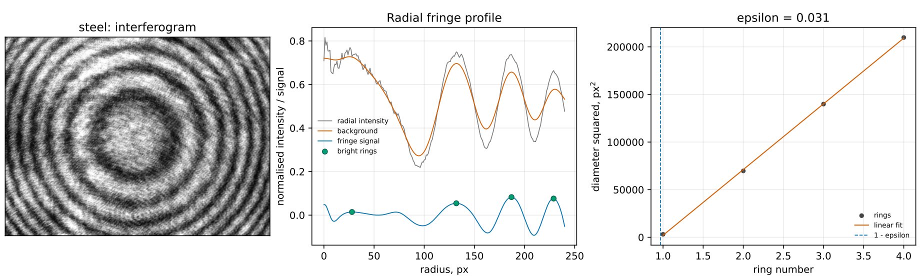

Figure S4: Steel single-image radial phase extraction. Same EFM procedure applied to a steel interferogram (640×480 px).

3.2. Sequence processing steps

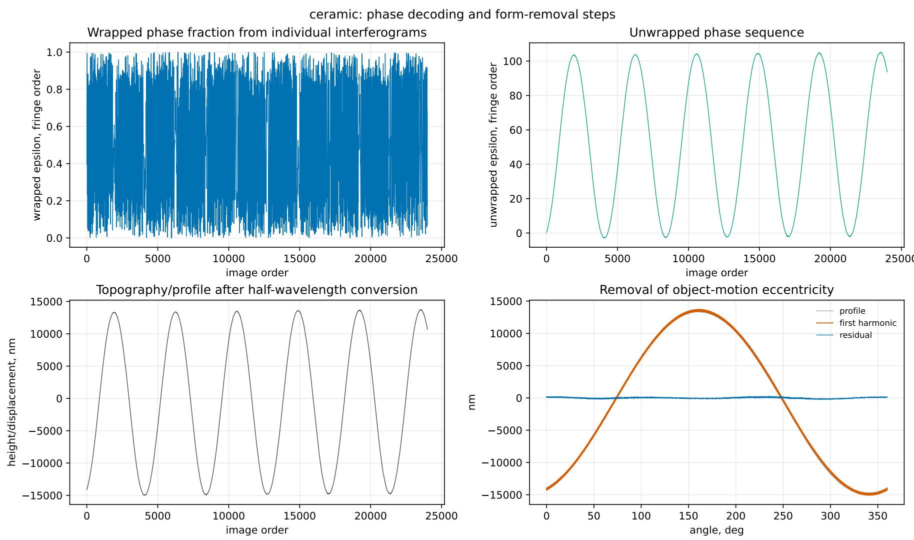

Figure S5: Ceramic sequence processing. (Top left) Wrapped excess fraction. (Top right) Unwrapped fringe order. (Bottom left) Nanometre height profile with LSCI reference circle and eccentricity fit. (Bottom right) Gaussian low-pass filtered residual (50% at 15 UPR, ISO 16610-21).

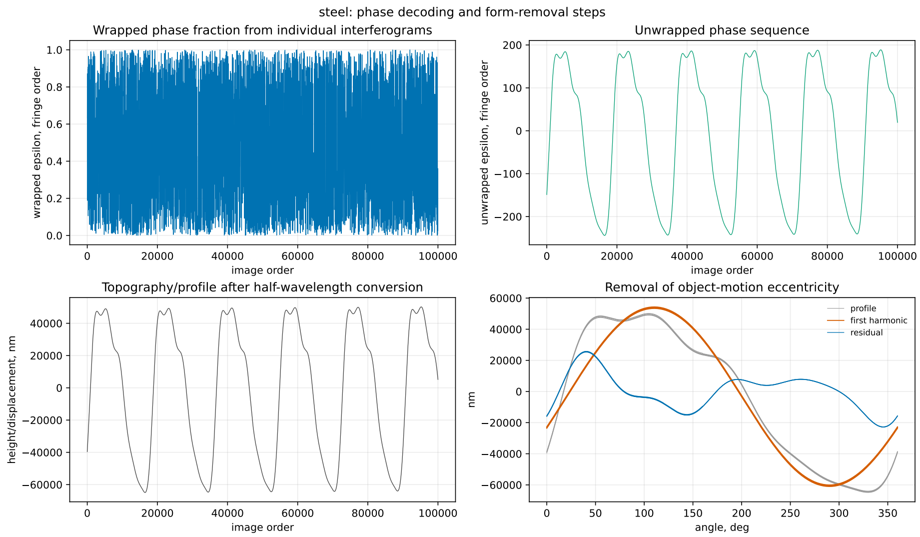

Figure S6: Steel sequence processing. (Top left) Wrapped excess fraction. (Top right) Unwrapped fringe order with linear drift. (Bottom left) Nanometre height profile after detrending. (Bottom right) Gaussian low-pass filtered residual.

4. Results — Descriptor Comparisons

4.1. Descriptor comparison and uncertainty

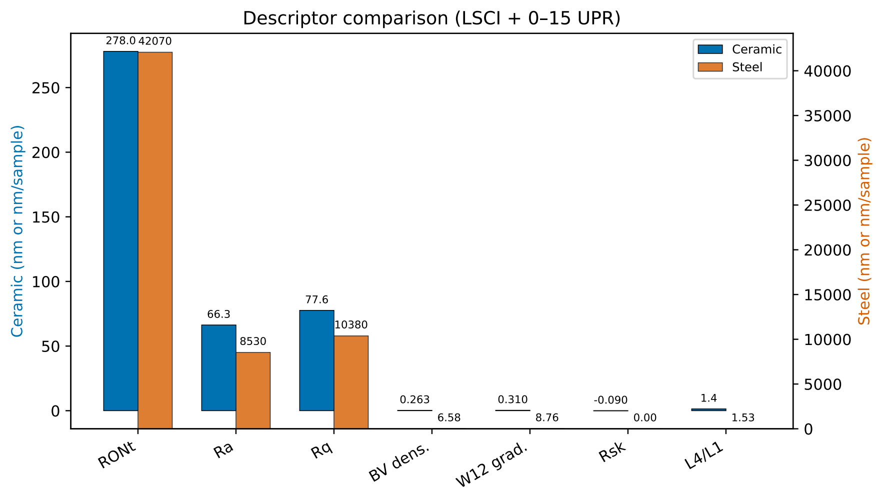

Figure S7: Comparison of ISO and gradient-based descriptors for both artefacts (bar charts). FN 111 and FN 101 values shown side by side. Error bars: 95% expanded within-run repeatability ($n = 5$).

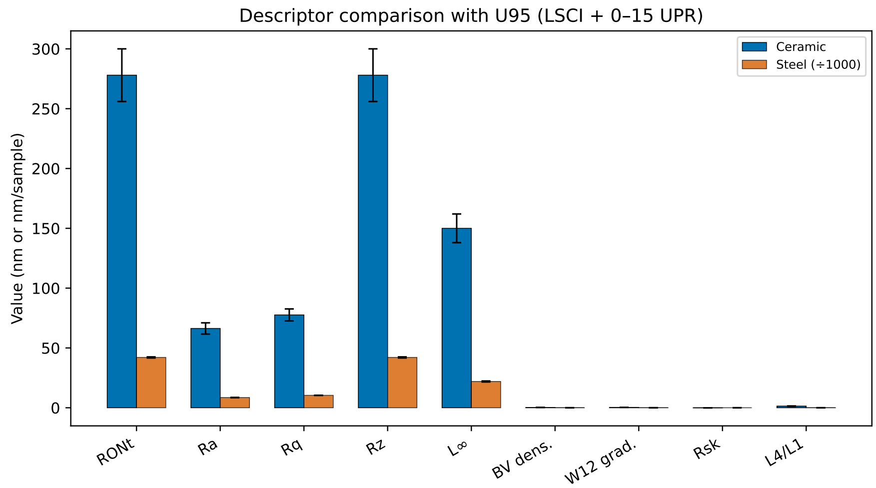

Figure S8: Uncertainty comparison: relative expanded uncertainty for each descriptor, colour-coded by descriptor family (ISO/profile vs. Banach/gradient), for both artefacts.

4.2. Profile reconstruction

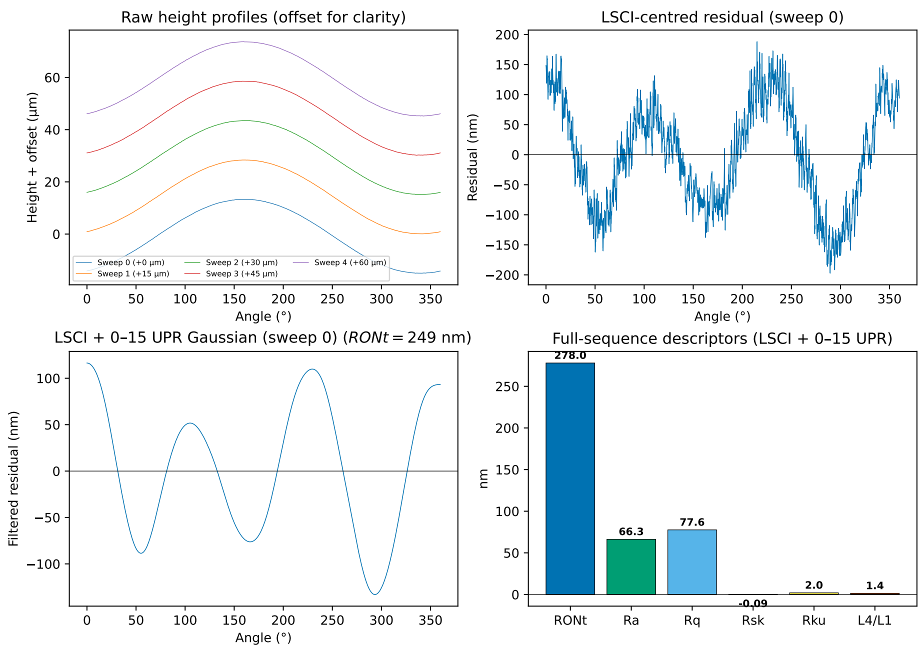

Figure S9: Ceramic profile reconstruction. (Top left) Raw height profiles from all five sweeps, vertically offset for clarity. (Top right) LSCI-centred residual for sweep 0. (Bottom left) Gaussian low-pass filtered residual (50% at 15 UPR) with $RONt = 278$ nm. (Bottom right) Full-sequence ISO and gradient-based descriptors.

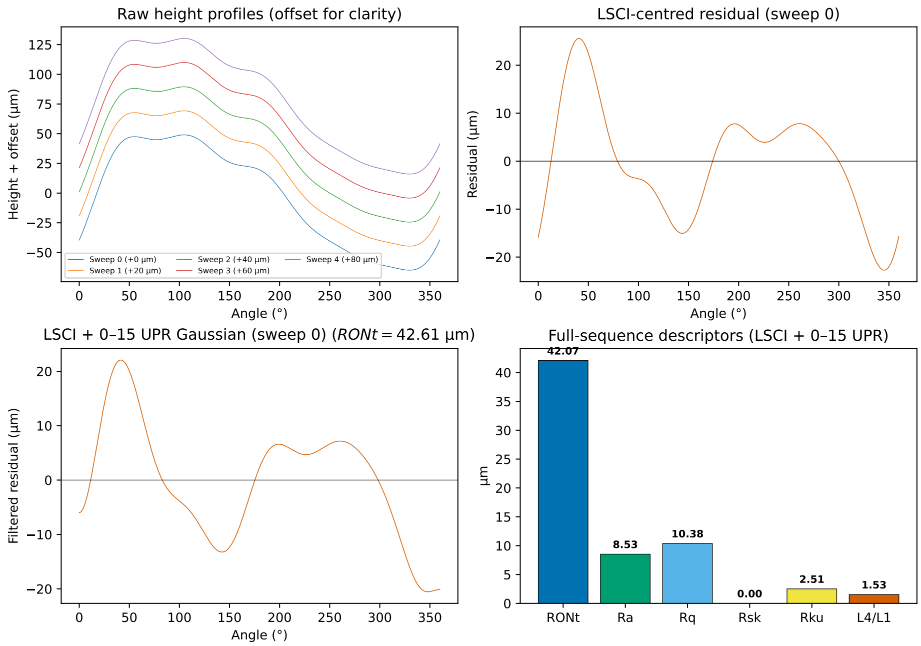

Figure S10: Steel profile reconstruction. (Top left) Detrended height profiles from all five sweeps, vertically offset. (Top right) LSCI-centred residual for sweep 0. (Bottom left) Gaussian low-pass filtered residual (50% at 15 UPR) with $RONt = 42.07$ μm. (Bottom right) Full-sequence ISO and gradient-based descriptors.

4.3. Steel roundness — polar representation

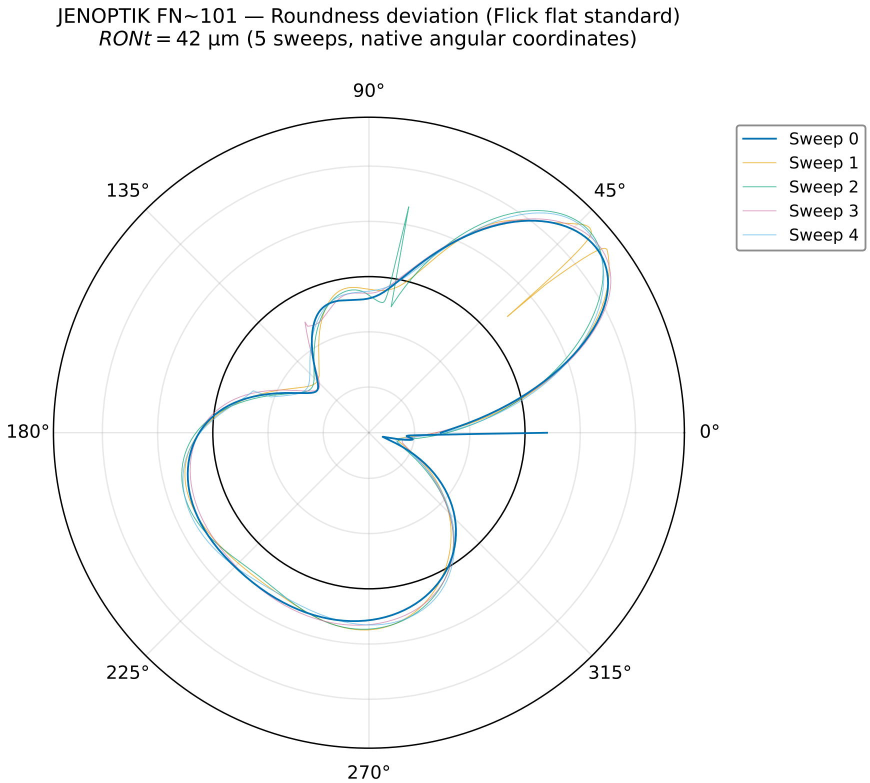

Figure S11: Steel roundness in polar coordinates (LSCI + Gaussian low-pass, 50% at 15 UPR). Five sweeps at native angular positions. The Flick flat appears at a shifted angular position in each sweep due to the 40° offset between successive 360° segments. $RONt = 42.07 \pm 0.48$ μm.

4.4. Per-sweep descriptor values

Table S2: Per-sweep descriptor values computed from the LSCI-centred residual. The Gaussian low-pass filter (50% at 15 UPR, ISO 16610-21) is applied after concatenating all five sweeps into a single sequence; per-sweep filtered values are therefore reported before filtering. The sweep-to-sweep variability pattern is preserved; absolute values are higher than the filtered means in Table 2 of the main manuscript. Each row reports the five individual sweep values followed by the mean ± expanded uncertainty ($U_{95} = t_{0.975,4}\, s/\sqrt{5}$, $t_{0.975,4} = 2.776$). Absolute values are higher than those in Table 2 of the main manuscript because the 15 UPR Gaussian low-pass filter has not been applied; the sweep-to-sweep variability pattern is preserved.

	FN 111 (ceramic)						FN 101 (steel)					
Descriptor	S0	S1	S2	S3	S4	Mean ± U_{95}	S0	S1	S2	S3	S4	Mean ± U_{95}
R_a	86.07	80.97	76.07	82.56	81.81	81.5 ± 4.5	9.368	9.376	9.398	9.371	9.344	9.371 ± 0.024
R_q	103.9	94.20	91.14	98.73	100.8	97.7 ± 6.3	11.54	11.56	11.59	11.58	11.55	11.564 ± 0.025
R_z	477.7	427.0	441.9	479.5	492.5	464 ± 35	48.26	48.48	48.49	48.49	48.49	48.44 ± 0.13
$RONt$	477.7	427.0	441.9	479.5	492.5	464 ± 35	48.26	48.48	48.49	48.49	48.49	48.44 ± 0.13
$RONq$	103.9	94.20	91.14	98.73	100.8	97.7 ± 6.3	11.54	11.56	11.59	11.58	11.55	11.564 ± 0.025
L^1 mean	86.07	80.97	76.07	82.56	81.81	81.5 ± 4.5	9.368	9.376	9.398	9.371	9.344	9.371 ± 0.024
L^2 RMS	103.9	94.20	91.14	98.73	100.8	97.7 ± 6.3	11.54	11.56	11.59	11.58	11.55	11.564 ± 0.025
L^∞	257.3	221.7	234.9	265.2	261.4	248 ± 23	25.50	25.69	25.68	25.65	25.63	25.629 ± 0.095
BV total	30752	31200	32567	32616	34197	32266 ± 1685	150.3	150.3	150.3	150.3	150.3	150.3383 ± 0.0083
BV density	7.119	7.222	7.539	7.550	7.916	7.47 ± 0.39	0.0084	0.0084	0.0084	0.0084	0.0084	0.0084 ± 0.0000
$W^{1,2}$ grad.	9.556	9.708	10.43	10.10	10.69	10.10 ± 0.59	0.0108	0.0108	0.0108	0.0108	0.0108	0.0108 ± 0.0000

5. Validation Analyses

5.1. Cross-instrument validation: optical versus tactile

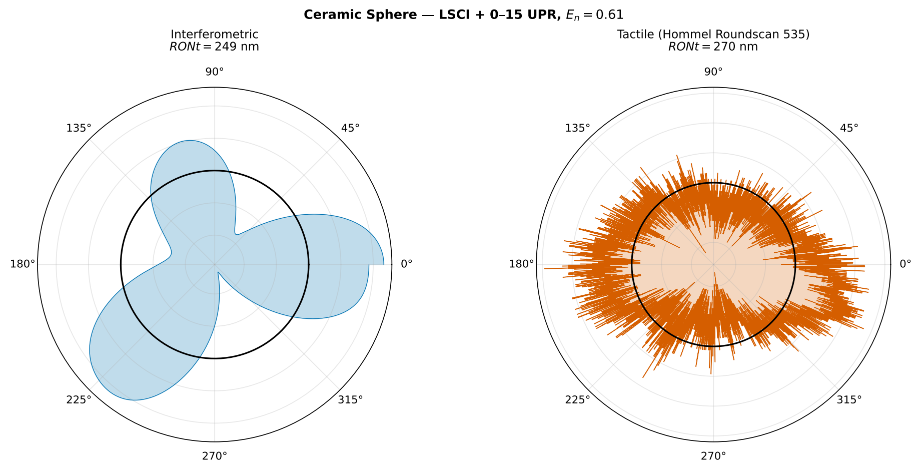

Figure S12: Polar comparison of optical (this work) and tactile reference (Hommel Roundscan 535) roundness profiles for the FN 111 standard. Both profiles are LSCI-centred and Gaussian low-pass filtered (50% at 15 UPR). The optical profile agrees with the tactile reference to within the combined uncertainty ($E_n = 0.61$).

5.2. Descriptor independence validation

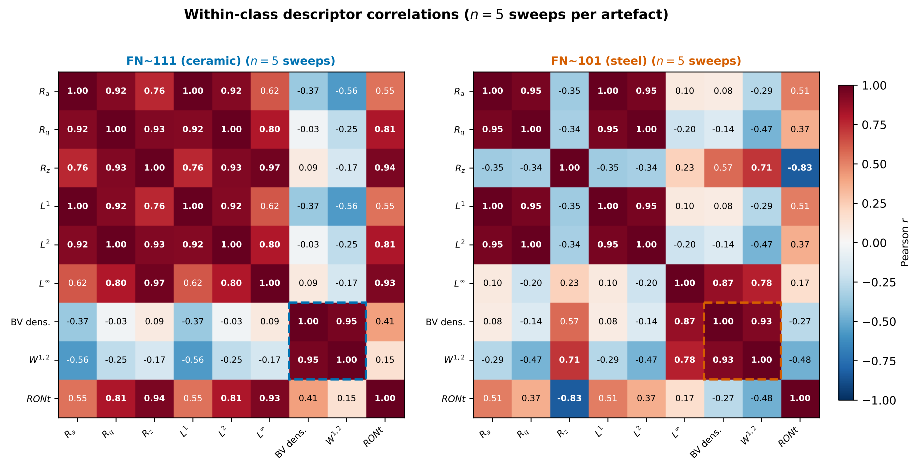

Figure S13: Within-class descriptor correlation matrices for FN 111 (left) and FN 101 (right). Dashed boxes highlight the Banach gradient-structure block.

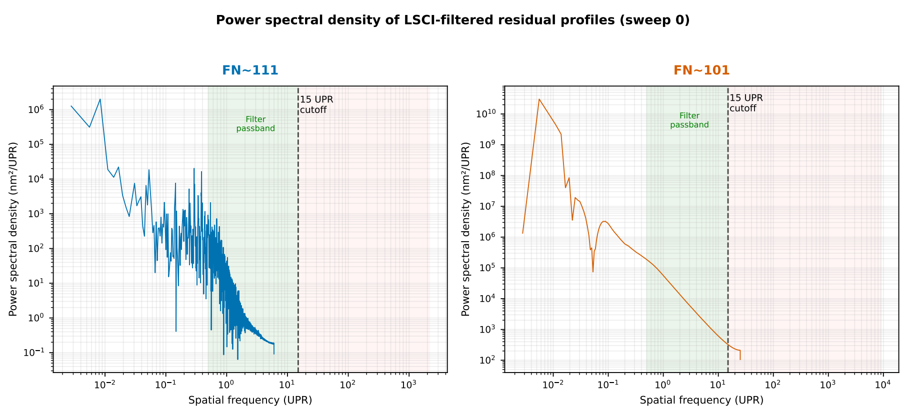

Figure S14: Power spectral density of LSCI-filtered residual profiles (sweep 0). Shaded region: Gaussian filter passband (0.5–15 UPR).

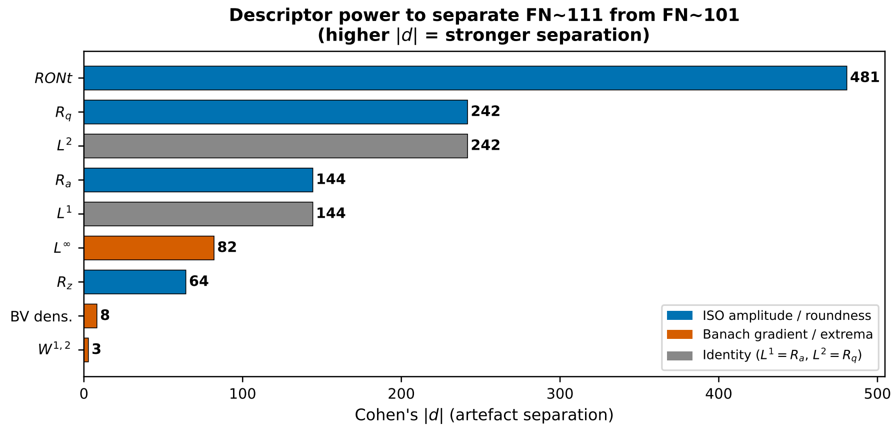

Figure S15: Cohen's $|d|$ effect size for separating FN 111 from FN 101. ISO amplitude/roundness descriptors compared with Banach gradient/extremum descriptors.

5.3. Critical descriptor assessment

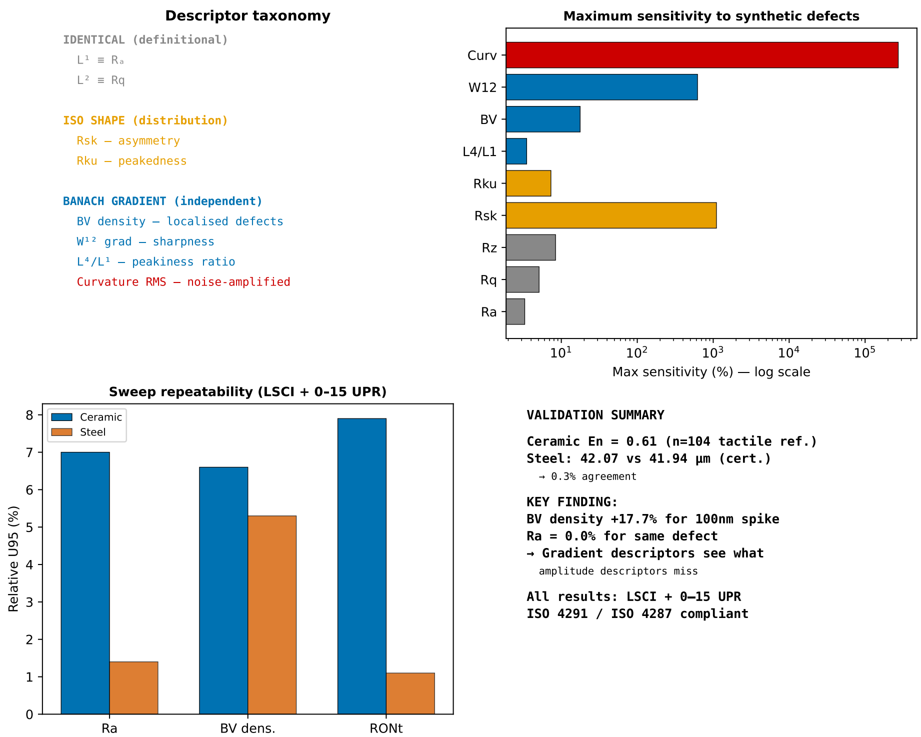

Figure S16: Critical assessment of gradient-based versus ISO descriptors. (Top left) Descriptor taxonomy. (Top right) Maximum sensitivity to synthetic defects. (Bottom left) Sweep-to-sweep stability. (Bottom right) Information-content summary.

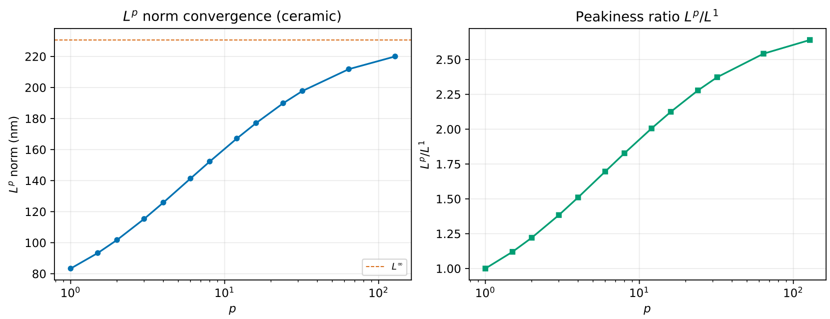

Figure S17: L^p norm convergence for the FN 111 residual profile (unfiltered). (Left) L^p as a function of p. (Right) Peakiness ratio L^p/L^1.

6. Sensitivity Analysis and Flick Flat Detection

6.1. Descriptor sensitivity to synthetic defects

Table S3: Sensitivity of ISO and gradient-based descriptors to synthetic profile defects. Values are percentage change from the unperturbed ceramic baseline (LSCI + Gaussian low-pass, 50% at 15 UPR; R_a = 61.9 nm, R_q = 71.0 nm). Bold entries indicate changes exceeding 10%. Defect details: Spike = single 100 nm point; Scratch = 5 × 50 nm depression; Waviness = 20 nm sinusoid (λ = 500 samples); Step = 30 nm offset; Noise = 5 nm RMS white noise. The baseline values (R_a = 61.9 nm, R_q = 71.0 nm) correspond to the unperturbed sweep-0 ceramic profile used for defect injection; they differ slightly from the five-sweep means in Table 2 of the main manuscript because the latter average all sweeps with the 40° angular offset between successive 360° segments.

Descriptor	Spike	Scratch	Waviness	Step	Noise
		Amplitude-averaging ISO parameters			
R_a	−0.0%	+0.1%	+3.2%	+3.3%	+0.1%
R_q	−0.0%	+0.1%	+3.3%	+5.1%	+0.2%
R_z (PV)	+0.0%	+0.0%	+7.0%	−2.6%	+8.4%
		ISO shape parameters			
R_{sk} (skewness)	−0.4%	+6.7%	**−168.5%**	**−1103.7%**	−0.1%
R_{ku} (kurtosis)	+0.0%	+0.1%	+2.4%	+7.3%	+0.4%
		Gradient-structure descriptors			
L^4/L^1 peakiness	+0.0%	+0.0%	+0.7%	+3.5%	+0.2%
BV density	**+17.7%**	+8.9%	**+11.7%**	+2.7%	**+2077.4%**
$W^{1,2}$ grad. seminorm	**+621.8%**	**+271.2%**	**+16.9%**	**+81.7%**	**+2260.9%**
Curvature RMS*	—	—	**+94%**	—	—

The results lead to five conclusions:

1. Amplitude-averaging ISO descriptors (R_a, R_q, R_z) show negligible sensitivity to localised defects. A single 100 nm spike changes R_a by 0.0%, R_q by 0.0%, and R_z by 0.0%.
2. ISO shape parameters respond to asymmetry but not to symmetric localised features. R_{sk} changes by −1104% for a step defect (strong asym-

metry), but only -0.4% for a symmetric spike. R_{ku} is insensitive to all tested defects.

3. Gradient-structure descriptors provide substantially higher sensitivity to localised defects than any single ISO 21920-2 (formerly ISO 4287) parameter. BV density increases by $+18\%$ for a spike, $+9\%$ for a scratch, and $+12\%$ for waviness. The $W^{1,2}$ gradient seminorm shows even larger relative changes ($+622\%$ for spike, $+271\%$ for scratch).
4. R_{sk} and BV density are complementary: R_{sk} detects asymmetry, BV density detects localised gradients regardless of symmetry. Their combination provides descriptive sensitivity beyond any single classical parameter.
5. Curvature RMS is strongly sampling-dependent, carrying a $(\Delta\alpha)^{-2}$ factor that amplifies single-point second differences for spike- and scratch-type defects; its percentage-change values for such defects are not practically interpretable and are omitted from the table (see footnote). Curvature RMS should be used only for same-instrument, same-sampling comparisons of smoothly varying profiles.

* Curvature RMS carries a $(\Delta\alpha)^{-2}$ sampling dependence that amplifies single-point second differences for spike- and scratch-type defects by factors exceeding 10^3; its percentage-change values for such defects are not practically interpretable and are omitted. The $+94\%$ change for the waviness defect reflects the sinusoidal curvature contribution and is meaningful for same-instrument comparisons of smoothly varying profiles.

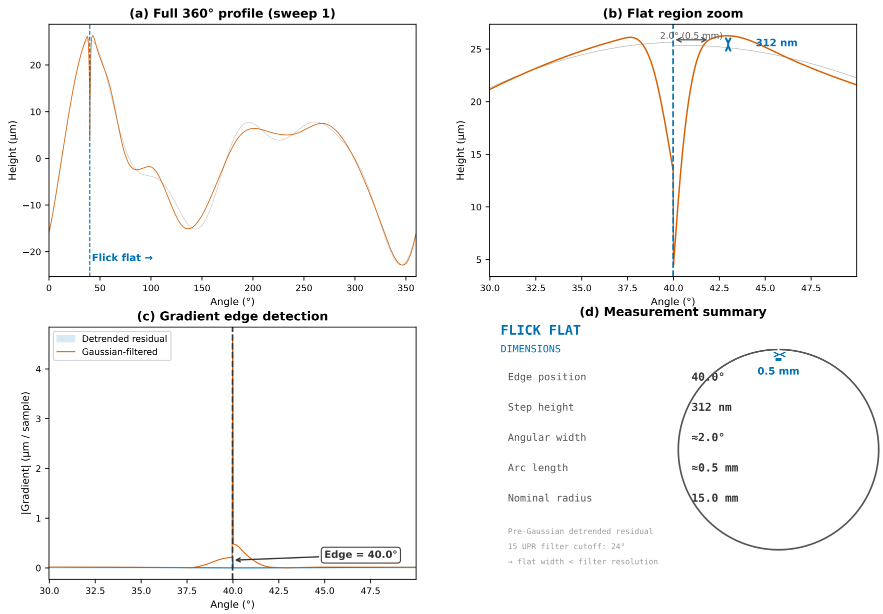

Figure S18: Flick flat measurement on the JENOPTIK magnification standard FN 101. Reconstructed profile across the Flick flat sector showing the characteristic step-like geometry, with the flat region and adjacent roundness profile labelled. $RONt = 42.07$ μm is dominated by the Flick flat depth.

7. Harmonic Decomposition

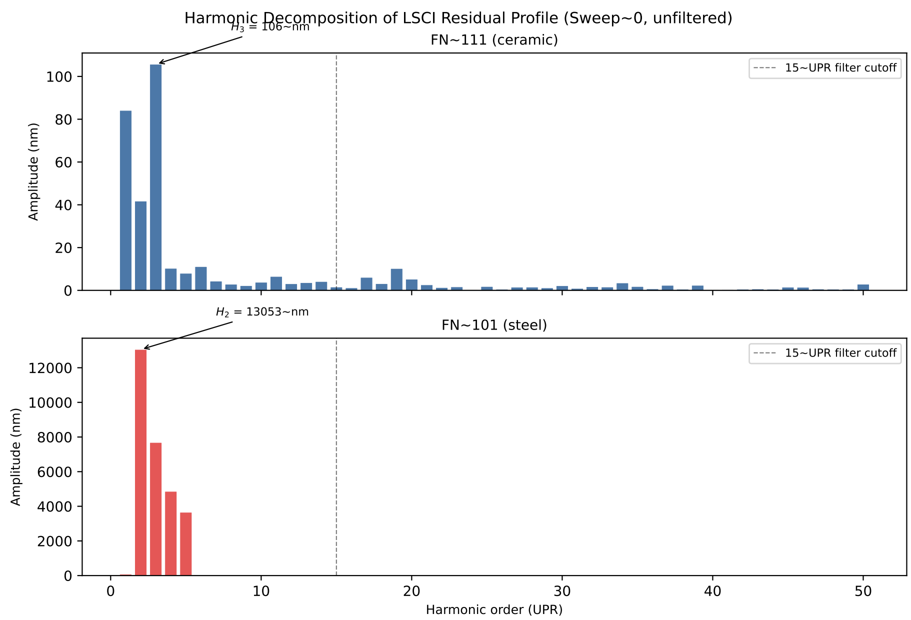

Figure S19: Harmonic decomposition of the LSCI residual profile (sweep 0, pre-Gaussian-filter). Dominant harmonics are annotated; the 15 UPR Gaussian filter cutoff (dashed line) attenuates harmonics above this order. The ceramic residual is dominated by the 2nd harmonic (H_2), consistent with the single-lobe form explaining the $RONt = R_z$ identity after filtering. The steel residual shows a broader harmonic distribution reflecting the Flick flat's step-like geometry.

References

[1] Newport Corporation, Precision Motion Control Catalog, Newport Corporation, 2018, (accessed 16 August 2026).
URL https://www.newport.com/p/URS100BCC